P510/1
PHYSICS
Paper 1
(Theory)
July/August 2026
3 hours

WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Advanced Certificate of Education

PHYSICS

Paper 1

3 hours

INSTRUCTIONS TO CANDIDATES:

- *This examination paper consists of two sections* **A** *and* **B.** *it has* **six** *items.*
- *Section* **A** *has* **two** *compulsory items.*
- *Section* **B** *has two parts;* **I** *and* **II.** *Respond to only* **one** *item from each part.*
- *Respond to* **four** *items in all.*
- *Any additional item(s) responded to, will* **not** *be scored.*
- *Responses to* **each** *item should start on a fresh page.*
- *Silent non-programmable scientific calculators may be used.*

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Turn Over

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SECTION A

Respond to **all** *the items in this section.*

Item 1

As part of preparations for the PAMOJA AFCON Tournament, the Uganda Cranes conducted a series of training sessions involving football skills, physical conditioning and transportation to their hill top training camp.

During a free-kick practice session, a striker places the ball 20.0 m directly in front of the centre of the goal. A defensive wall stands 9.0 m from the ball and is 2.0 m high, while the crossbar is 2.44 m above the ground. The sticker intends to kick the ball so that it just clears the wall before entering the goal. Air resistance during the kick is neglected.

In another training drill, the same football of mass 0.43 kg is released from rest from a height of 30.0 m above the ground. As it falls, the ball experiences air resistance and eventually reaches terminal velocity before striking the ground. The ball has a circumference of 68 cm and an internal pressure of $1.2 \times 105\ p_a$.

The team's training camp is located at the top of a hill inclined at 30^0 to the horizontal. A team bus of mass 2500 kg starts from rest and accelerates uniformly at 5.0 ms^{-2} for 10 s while travelling up the slope, neglecting friction at the tyres.

The players appreciated the training activities and transportation but were unable to explain the physics involved in the different situations.

Task

(a) Analyse the information provided to explain the mechanics involved in football training activities and the team's journey to the training camp.

(b) Evaluate the performance of the football and the team bus using appropriate principles of mechanics and recommend suitable measures to improve their performance.

Item 2

At a national research and energy innovation center, a group of students visits laboratories studying nuclear energy, radiation detection and electron behavior in electric and magnetic fields. The visit aims to expose students to how atomic and electronic principles are applied in modern science.

In the nuclear research section, scientists explain how the stability of nuclei depends on binding energy per nucleon and how energy is released in nuclear processes.

The students observe a graphics relationship between binding energy per nucleon and mass number.

In the radiation laboratory, a Geiger-Muller tube is used to detect radiation from a thorium sample. The scientists note that counts are still recorded even when no radioactive source is present. The thorium-230 of mass 1g and half-life 75,400 years was detected by the machine.

In another experiment, students study the emission of electrons from metal surfaces when illuminated by light and observe how X-rays are produced in a vacuum tube.

Finally in the electron dynamics laboratory, they observe the motion of electrons in an electric fields and how their specific charge can be determined experimentally.

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Task

(a) Analyse the information provided to explain the observation made in the nuclear radiation, photoelectric X-ray and electron dynamics experiments using appropriate principles and carrying out relevant calculations.

(b) Describe the motion of electrons in electric and magnetic fields and explain how their specific charge is determined experimentally, highlighting its significance in modern applications.

SECTION B

Part 1

Respond to **one** *item from this part.*

Item 3

A biconvex silica lens ($n = 1.45$ at 1064 nm) is used to focus a 100W continuous - wave laser. Because the laser power is high, the lens absorbs heat, changing its refractive index and shape (thermal lensing). The lens is placed in a custom continuous flow calorimeter to determine the exact focal length under operation by monitoring its heat absorption.

Support material:

- The lens is manufactured as biconvex, with radius of curvature $R_1 = +50$ mm
- and $R_2 = -50$ mm.
- The lens is submerged in the flow chamber surrounded by a specialized cooling oil of refractive index $n_{oil} = 1.30$.
- The cooling oil flows around the lens at a constant rate $m = 0.05\ kgs^{-1}$.
- The inlet temperature $T_1 = 20.0^0C$ and outlet temperature $T_2 = 22.5^0C$ are measured.
- The specific heat capacity of oil is $C_{oil} = 2100\ Jkg^{-1}k^{-1}$.
- Heat loss to the environment is negligible.
- The refractive index of the lens material increases to $n = 1.48$ due to thermal heating.

Task:

(a) Analyse the information provided to determine the focal length of the lens and the energy absorbed by it under operation.

(b) Explain the operation of the continuous flows calorimeter and how it is used to measure heat absorbed by the lens.

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Item 4

A company is planning to start a project of wave power around Lake Victoria. The company project manager approached an engineer. The engineer explained that the available wave power device can only generate electricity if the wave amplitude is at least 0.155 m. The manager therefore needed to determine whether the lake waves are suitable for the project.

The lake wave is modelled by the equation $y = A \sin\left(\frac{2\pi t}{T} - kx\right)$, where T is the period of the wave and k is the wave number, y is the displacement of one wave in meters from undisturbed surface and x is the distance moved by the wave. For future research purposes the manager would like to have a record of the wave equation of the water waves.

Because the direct measurement of the moving wave is difficult, the manger decided to use a screen method based on reflection of light to determine amplitude.

A narrow laser beam is directed onto the water surface. The reflected beam falls on a vertical screen placed 100 m way. When light is reflected from the crest of the wave it reaches the mean position O on the screen. When light is reflected from the wave surface at it's greatest positive slope, the reflected sports shifts 0.8 m above the mean position O. The students need to use knowledge of light reflection on rotated reflecting surface to determine the inclination of greatest slope of the wave.

To check the accuracy of the manager's results, engineer used as second method. A converging lens is placed in the path of the reflected beam so that the light is focused into an optical fiber. The optical fiber carries the light to a photo detector and the electronically recorded signal is compared with the manager's screen measurements. The manager does not know how the optical fiber transmits the light and why a convex lens was used in this option by the engineers.

Support material

- The angle of inclination of greatest slope of the wave is equal to the maximum slope of the wave.
- Wave length $\lambda = 25.0$ m and period $T = 10.0$ S

Task:

(a) Analyse the information provided to help the manager determine whether the lake waves can support power generation and establish the wave equation.

(b) Discuss the accuracy of the optical measurement method and explain how the second method using a lens and optical fiber improves reliability.

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PART II

Respond to **one** *item from this part.*

Item 5

At an electronics and material testing laboratory, technicians are designing capacitors for use in power supplies, radio circuits and energy storage units.

To ensure safe and efficient operation, they must understand how charge, voltage, geometry and insulating materials affect capacitor or behaviour.

In one test a metallic spherical of radius 7.5 cm is connected to a high voltage source of 9.0 V to study how much electric charge it can safely hold.

In another experiment, two large metal spheres are mounted facing each other and their separation is varied while the resulting electrical properties are observed.

Engineers also investigate how inserting different materials between capacitor plates affects charge storage and how a ballistic galvanometer can be used to determine electrical properties of insulating materials with high precision.

In the same laboratory, a technician of physics sets up several test circuits supplied by a stable D.C power source.

The aim is to verify fundamental electrical laws, calibrate measuring instruments and evaluate the performance of power sources used in students experiments.

In the first set up, a uniform metallic conductor is connected in series with a variable resistor, an ammeter and a switch. A voltmeter is also connected across a uniform conductor. The technician varies the current through the conductor while carefully monitoring the potential difference across it, ensuring that the temperature of the conductor remains nearly constant throughout the test.

The technician also evaluates the efficiency of a practical power circuit consisting of a battery of emf 20 V and internal resistance 4 Ω supplying a load resistor of resistance 10 Ω.

The analysis helps to determine how much of the chemical energy in the battery is usefully converted into electrical energy in the external circuit.

Finally engineers investigate why delicate electronic equipments are frequently destroyed during operation especially when they are near highly charged machinery.

Support material.

Area of capacitor plates = 5.0 cm^2

Separation of the plates = 2.0 mm

P.d across the plates = 6.0 V

Permittivity of free space = 8.85 x 10^{-12} Fm^{-1}

Task

(a) Analyse the information provided to determine the charge stored on the capacitor system, evaluate the efficiency of the battery using appropriate electrical principles and relevant calculations.

(b) Explain how capacitance depends on geometry and dielectric materials, use of a ballistic galvanometer in material testing and account for equipment damage near highly charged systems, including prevention measures.

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Item 6

A large agricultural irrigation farm operates an electrically driven water pump used to distribute water across different sections of land.

The pump is powered by a steady alternating current supply generated on site using a rotating – coil generator driven by a diesel engine. The generator has 500 turns in its coil, rotating in a uniform magnetic field of strength 0.30T.

Each coil has an area of 0.020 m^2 and the generator rotates at a steady frequency of 50 Hz. One farmer wonders how a magnet and wire coiled together give electricity.

To improve stability and protect the motor of the pump, the pump circuit includes;

- A series inductor of inductance 0.60 H
- A capacitor of capacitance 60 μF
- A motor resistor of resistance 20 Ω.

They want to add a logic base control system that ensures the pump operates only when water demand is required and the electrical conditions are safe.

However, the farmers report that the pump operates less efficiently than expected with reduced useful mechanical output compared to electrical energy supplied. Engineers suspect that there is an unknown resistance affecting how the motor draw current from the source.

Task:

(a) Analyse the information provided to determine the performance of the pump system and advise the engineer how to improve its efficiency.

(b) Explain how electrical energy is generated by the rotating coil generator and propose a suitable logical control strategy for safe operation of the pump.

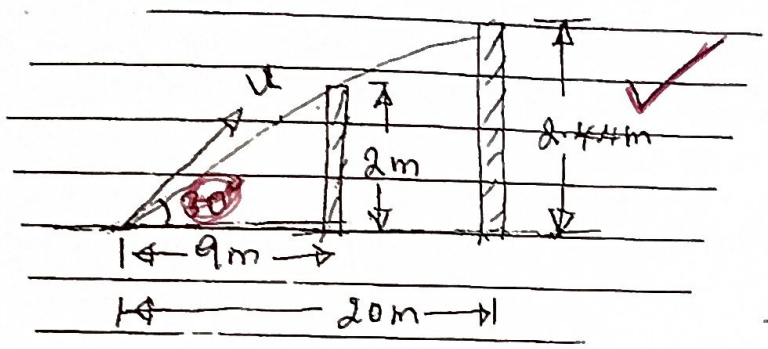
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Substitution - 02
θ - 01
U - 01
Units
diagram - 01
Total - 06

Item 1
(a)



$U = ?$
$Q = 30^0$
$X = 9.0m$
$Y = 2.0m$

$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

let $\tan\theta = A$, $g = B$

$2 = 9A - 81B$ — (i) ✓

$2.44 = 20A - 400B$ — (ii) ✓

$y = Ax - Bx^2$

$A = 0.3042$

$B = 0.009111$

$\theta = 16.9°$ ✓

$U = 24.26 ms^{-1}$ ✓

05

Using

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$$

$$2 = 9\tan 30^0 - \frac{(9.81)(9.0)^2}{2u^2\cos^2 30^0}$$

$$2 = 9(0.0577) - \frac{794.6}{1.5u^2}$$

$$u^2 = 165.907$$

$$u = 12.88 m/s$$

Conclusion: ✓

At $u = 12.88 m/s$

When the ball reaches goal post $x = 20\ m$

01

So

$$y = 20\tan 30^0 - \frac{(9.81)(20)^2}{2(165.907)\cos^2 30}$$

$$= 11.547 - 15.767$$

$$= -4.22m$$

This means that the ball landed before it reached the goal posts. The stricker did n't score a goal and Uganda failed to qualify for the round of 16 at PAMOJA AFCON tournament.

03

However if the ball is kicked at angle of 45^0 ;

$$y = 9\tan 45^0 - \frac{(9.81)(9)^2}{2u^2\cos^2 45^0}$$

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- explanation of terminal - 04 - Explain terminal velocity ✓

Radius - 01

Substitution - 01

$F_V = 4.1476N$ - 01

Subst. - $6\pi\eta V_0$ - 01

V_0 - 01

Volume - 01

Total - 10

$U + F_V = mg$ ✓ (04)

$\frac{\rho}{2} \times \frac{4}{3}\pi R^3 g + F_V = mg$ ✓

$1.25 \times \frac{4}{3}\pi \left(\frac{68\times10^{-2}}{2\pi}\right)^3 \times 9.81 + F_V = 0.43 \times 9.81$ ✓

$F_V = 4.1476N$ ✓

$F_V = 6\pi\eta V_0 \rho$

$4.1476 = 6\pi(1.8\times10^{-5}) V_0 \times \left(\frac{68\times10^{-2}}{2\pi}\right)$

$V_0 = 1.13 \times 10^5 ms^{-1}$ ✓

$R = \frac{c}{2\pi} = \frac{68\times10^{-2}}{2\pi}$

$R = 0.1082m$ ✓

$2 = 9 - \frac{794.61}{u^2}$

$u^2 = 118.515$

$u = 10.654 ms^{-1}$

At this speed

$y = 9\tan 45^0 - \frac{9.81(9)^2}{2(113.515)Cos^2 45}$

$= 2m$

Volume = $0.00530977m^3$ from

This means that at 45^0, the stricker would score from the free kick and Uganda would qualify.

03

V - 02 ($50ms^{-2}$)

F_d - 02 (24762.5N)

Recommendation - 03 Any 3

Work done - 02

Total - 09

03

Car motion

Using $V^2 = U^2 + 2@s$

$(30)^2 = 2@(300)$

$@ = \frac{900}{600}$

$@ = 1.5ms^{-1}$

$V = u + at$

$= 0 + (5\times10)$

$= 50ms^{-1}$ ✓

Assuming the car has Q mass of 1000kg

$F = ma$

$= 1000 \times 1.5$

$= 1500N$

distance $s = ut + \frac{1}{2}at^2$

$s = \frac{1}{2}(5)(10)^2$

$= 250m$

$F_d = ma + mg\sin 30$

$= 2500(5) + 2500(9.8)(0.5)$

$F_d = 24762.5N$

Work done $= 24750 \times 250$

$= 6190626J$

05

Work one $W = \frac{1}{2}mv^2 = \frac{1}{2}(1000)(900)$

$= 4.5 \times 10^5 J$

For the drone propeller

Angular speed $w = 2\pi f$

$= 2\pi \times 40$

$= 80\pi\ rad\ s^{-1}$

Centripetal Acceleration

$@ = w^2 r$

$= (0\pi)^2 \times 0.20$

$= 1280\pi^2\ ms^{-2}$

01

(a) Improving the striker's performance.

The sticker should aim at kicking the ball at an angle of 45^0 and ensure sufficient initial speed.

02

For efficient transportation ensure the car engine provides sufficient energy and ensure the tyres grip the road to prevent skidding.

Total = 25

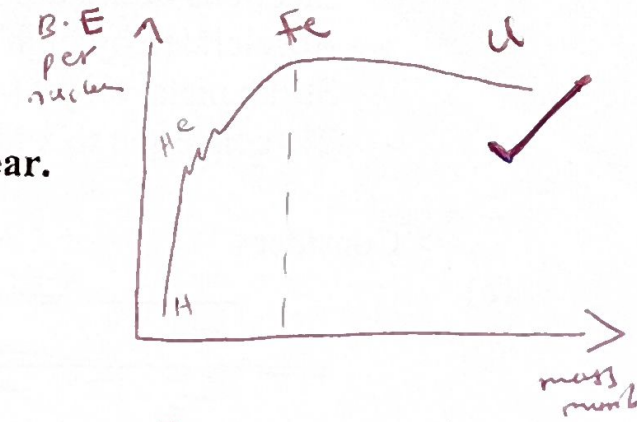
- During a nuclear (reaction) process,

Graph - 01
Explanation - 03
Energy release - 03 → 03
GMT - diagram (01), exp - 01
Background radiation - 01
Activity - 01

em 2

(b) In experiment one;
Binding energy per nuclear is the energy required to remove a nuclear from the nucleus.

i.e BE per nuclear = $\frac{\text{Total binding energy}}{A}$



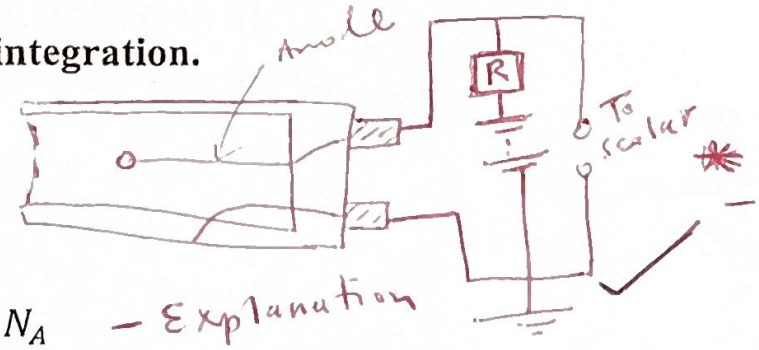
Variation of binding energy permition with that nuclear.
- Increases rapidly for lighter nuclear.
- It peaks around iron (A ≈ 56)
- Decreases slowly for heavy nuclear.

Energy released in fission and fusion.
- Lighter nuclear continue to form heavier nuclear.
- Products have higher binding energy per nuclear (move stable)

- For small nuclei of mass number < 56, the B·E per nucleon ↑ies rapidly and those close to 56 become more stable
- For larger nuclei, B·E per nucleon reduces gradually with ↑se in mass number.
- The stability of nuclei increases

In experiment two.
Geiger nuclear operation;
- Radiations enter the tube through nuclear window.
- Ionize gases (electrons accelerated to a node)
- Avalanche effect produce pulse.
- Pulse counted by the rate meter.

For back ground radiation.
- Counts recorded in absence of source.
- Due to cosmic rays and environmental radiations.



- Explanation

Rate of disintegration.
- $A = \lambda N$
- $\lambda = \frac{In2}{T_{\frac{1}{2}}}$
- $N = \frac{m}{M} \times N_A$

$M = 1g$ $M = 230$ $N_A = 6.02 \times 10^{23}$

$N = \frac{1}{230} \times 6.02 \times 10^{23}$

$\approx 2.62 \times 10^{21}$

$T_{\frac{1}{2}} = 7600 \times 3.65 \times 24 \times 3600 \simeq 2.40 \times 10^{12}$ s

$\lambda = \frac{0.693}{2.40 \times 10^{11}} \simeq 2.89 \times 10^{-13}$ s^{-1}

$A = \lambda N$

$= (2.89 \times 10^{-13})(2.62 \times 10^{21})$

$A = 7.62 \times 10^{8}$ Bq

Energy release

- During nuclear fusion, lighter nuclei combine to form a heavier stable nucleus of higher B·E per nucleon.
- The difference in the mass accounts for energy released.

→ For A > 56, heavy nucleus splits to form lighter nuclei that are more stable.
→ Difference in mass also accounts for energy released.

05

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How x-rays - 03,
photo-electric - 01
Electromagnetic - 02
J.J thompson - 03

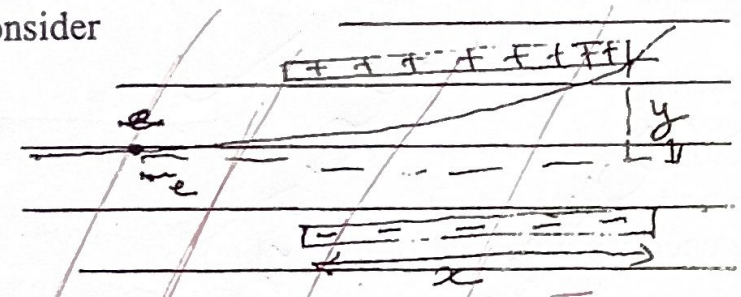
Experiment 3

X – ray production. *(Diagram 02)*

- Elections omitted from the cat bode (thermionic emission)
- Accelerated by high voltage
- Strike metal target (anode)
- 2% converted to x rays and the ret heat energy.

05

(b) Consider



In electric field, electrons experience an electric force and describe a parabolic path.

In a magnetic field, electrons experience a magnetic force and describe a circular path whose direction is perpendicular to the velocity of the electron

Horizontal motion

$S = x$ $\quad a = o\ m5^{-2}$

$u_x = u$

t = 1

From $s = ut + \frac{1}{2}at^2$

$X = ut$

$t = {}^{x}/_{u}$

Vertical nation

$\mathcal{U} = o$ $\quad F_e = ma$

$5^y = y$ $\quad ma = eE$

$A = \frac{eE}{m}$ $\quad a = \frac{eE}{m}$

$y = \frac{1}{2}at^2$

$y = \frac{1}{2}\frac{eE}{m}\left(\frac{x^2}{u^2}\right)$

$y = kx^2$ which in parabolic part

Use of J.J Thomson's experiment in deterring specific change.

- Abeam of electrons is subjected to across field. (Magnetic and electric field)
- At equiriblium

  Magnetic force = Electric force.

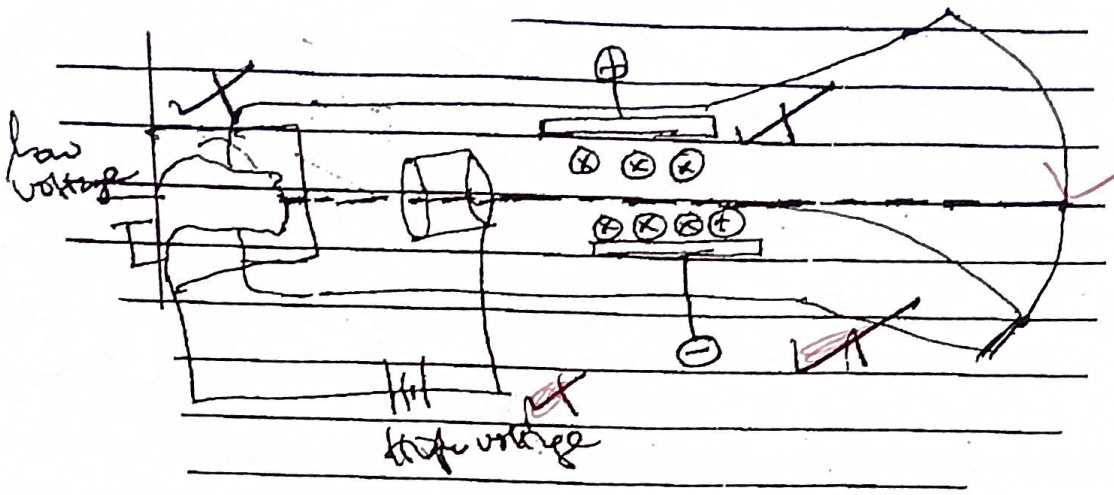
$$Beu = cE \quad \ldots\ldots\ldots\ldots\ldots \text{(i)}$$

$$\mathcal{U} = {}^{E}/_{\beta}$$

$$\text{Also } ev = \tfrac{1}{2}mu^2 \quad \ldots\ldots\ldots\ldots\ldots\text{(ii)}$$

From which

$${}^{e}/_{m} = \tfrac{1}{2}\left[\frac{E^2}{B^2}\right]$$ which gives the specific change of an election.



Accept the five beem experiment.

Application (significance) — CRT, mass spectrometer

05

Part I

Item 3

(a)

Case A

The lens makes formula is

$$\frac{1}{f} = \left(\frac{n\ lens}{nmed} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{fA} = \left(\frac{1.45}{1.80} - 1\right)\left(\frac{1}{50} - \frac{1}{-50}\right)$$

$$\frac{1}{fA} = (1.115 - 1)\left(\frac{2}{50}\right)$$

$$\frac{1}{fA} = \simeq 217.4\ mm$$

$f = 55.6mm$

Case B

Heated lens (nlens = 1.48, n med = 1.30)

$R_1 = 50, R_2 = -50$

$$\frac{1}{fB} = \left(\frac{1.48}{1.30} - 1\right)\left(\frac{1}{50} - \frac{1}{-50}\right)$$

$$\frac{1}{fB} = (1.138 - 1) \times 0.04$$

$$\frac{1}{fB} = 181.2mm.$$

Conclusion: Heating the lens changes the focal length from 217.4mm to 181.2mm.

The neat gained by the oil (Qabsorbed) is given by:

$Qabs = MC\Delta T$

$\Delta T = T_2 - T_1$

$\Delta T = 22.5 - 20.0$

$= 2.5k$

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$Qabs = (0.05) \times (2100) \times 2.5$

$Qabs = 262.5W$

Conclusion: This 262.5W represents the laser energy absorbed by the lens (due to imperfect transmission) and transferred to the oil.

(b) Continuous how calorimeter method:

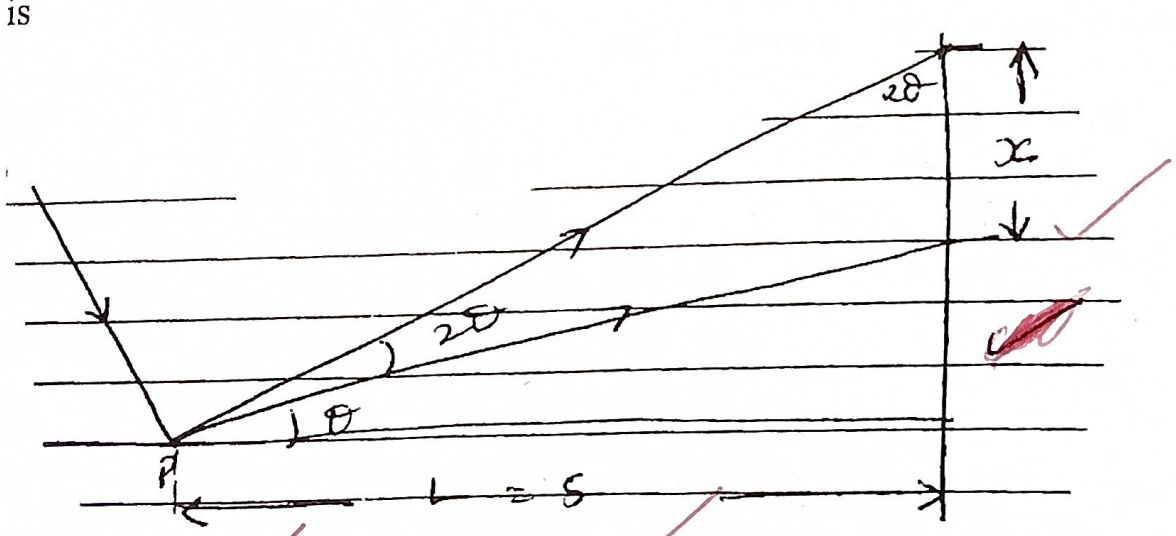
- Steady flow: A liquid (fluid) is passed through a tube or calorimeter vessel at a constant controlled rate.
- In let temperature check (T_1): A thermometer measures the incoming fluid temperature.
- Heat transfer: The liquid or fluid passes through a heater coil, absorbing the heat generated.
- Outlet temperature check (T_2): A second thermometer measures the outgoing warmer fluid.
- Steady state determination: the experimenter waits for the outlet temperature to stop changing, ensuring that all surrounding insulation has heated up and a "steady state" is achieved.
- Data collection and calculation: $C = \dfrac{V_2 I_2 - V_1 I_1}{(m_2 - m_1)(T_2 - T_1)}$

If M – is the mas flow rate.
ΔT – temperature rise.
C – Specific heat capacity.
$Q = MC\Delta T$

Total = 25

Item 4

(a) When the reflecting surface of water is filled through a small angle Q the reflected ray is



$\theta = \frac{x}{2l} = \frac{0.8}{2\times10} = \frac{0.8}{20} = 0.04\ radias$

Wave number $k = \frac{2\pi}{\lambda} = \frac{2\pi}{25} = 0.25$

Amputide $A = \frac{\omega}{k} = \frac{0.045}{0.25} = 0.16m$

The device can generate power if A > 0.155m since A = 0.16m > 0.155m, The project is supported by the lake wave.

$T = 10s$, $f = \frac{1}{10} = 0.1Hz$

The wave equation

$T = 10.0s \quad f = \frac{1}{10} = 0.1Hz$

Using $y = A \sin(2\pi ft - kx)$

$y = 0.16 \sin(2\pi(0.1)t - \frac{2\pi}{25})$

$y = 0.16 \sin(2\pi t - \frac{2\pi}{25})$

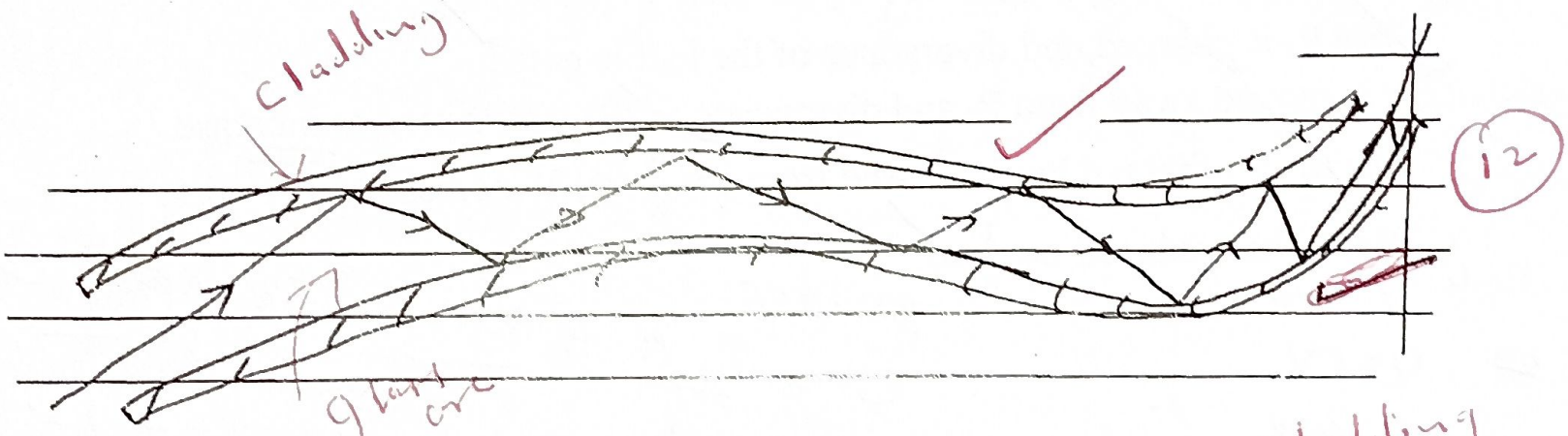
$y = 0.016 \sin[2\pi(0.1)t - 0.25x]$

$y = 0.016 \sin(0.2\pi t - 0.25x)m$

14

(b) The reflected beam from the wave may ~~speed~~ spread. The convex lens the collects the light and converges them. The beam is then focused into the small opening of the optical fiber. Almost all the reflected with is focused in the optical fiber this improves signal strength.

Optical fiber signal transmission



Optical fiber is made of a ~~cave~~ core that has more refractive index than the ~~clotting~~ cladding light is directed at the boundary between the core and the ~~clatting~~ cladding at an angle greater than critical angle.

The light is totally internally reflected multiple total internal reflection without loss of intensity.

This method is accurate because it reduces human energy improves sensitivity.

11

Item 5

(a) Capacitance is the ratio of Magnitude of charge on either mate of the capacitor to the p.d across its plates.

$C = 4\pi\varepsilon_{0r}$

$= 4\pi \times 8.85 \times 10^{-12} \times 7.5 \times 10^{2}$

$= 8.34\ 10^{-2}F \quad 8.34 \times 10^{12} F = 8.34 \mu F$

$Q = CV$

$= 8.34 \times 10^{-12} \times 9 \times 10^{3}$

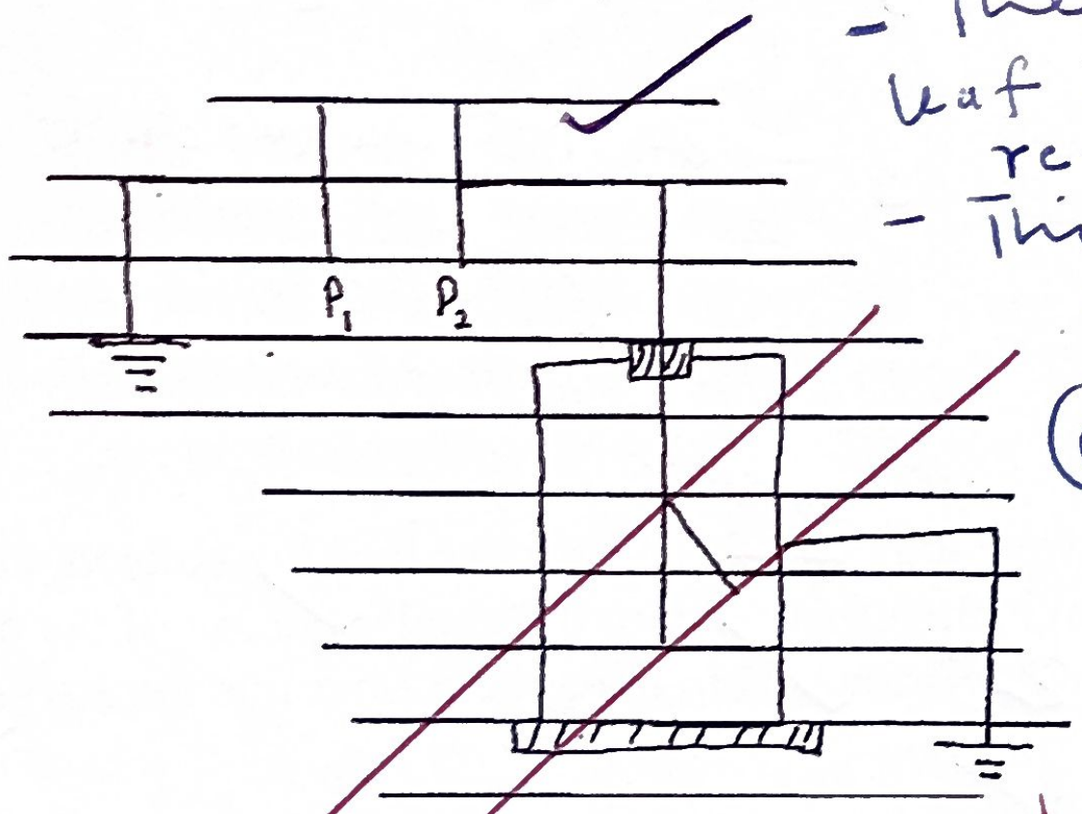
$= 7.5 \times 10^{-8}C \quad 7.51 \times 10^{-11} C$

- keeping area and separation of plates constant, a capacitor is filled with a dielectric material.
- The divergence of the leaf is noted and seen to reduce.
- This means p.d has reduced and capacitance has increased

(03)

$C \propto \frac{1}{V}$, C ↑ as it is dielectric.

(b)



(05)

P_1 and P_2 are metal plates. P_1 is earthed.

Plate P_2 is charged and divergence of the leaf is noted.

P_1 is moved away from P_2 and divergence of the leaf is seen to increase.

This means that p.d has increased from $c = \frac{Q}{V}$ and V increased.

Then C has reduced $c \propto \frac{1}{d}$

For the parallel plate capacitor

$C = \frac{\varepsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 5 \times 10^{-4}}{2 \times 10^{-3}}$

$C = 2.21 \times 10^{-12} F \ (2.21 pF)$

(04)

(a) $Q = CV$

$= \frac{\varepsilon_0 AV}{d}$

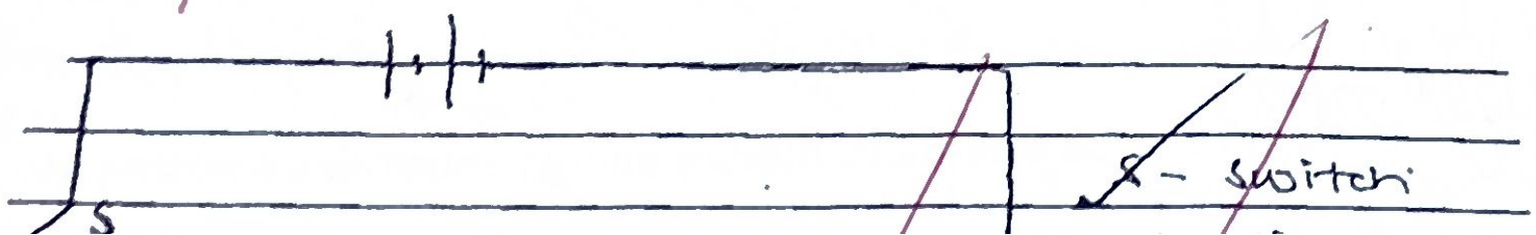
$= \frac{8.85 \times 10^{12} \times 5 \times 10^{-4} \times 6}{2 \times 10^{-3}}$

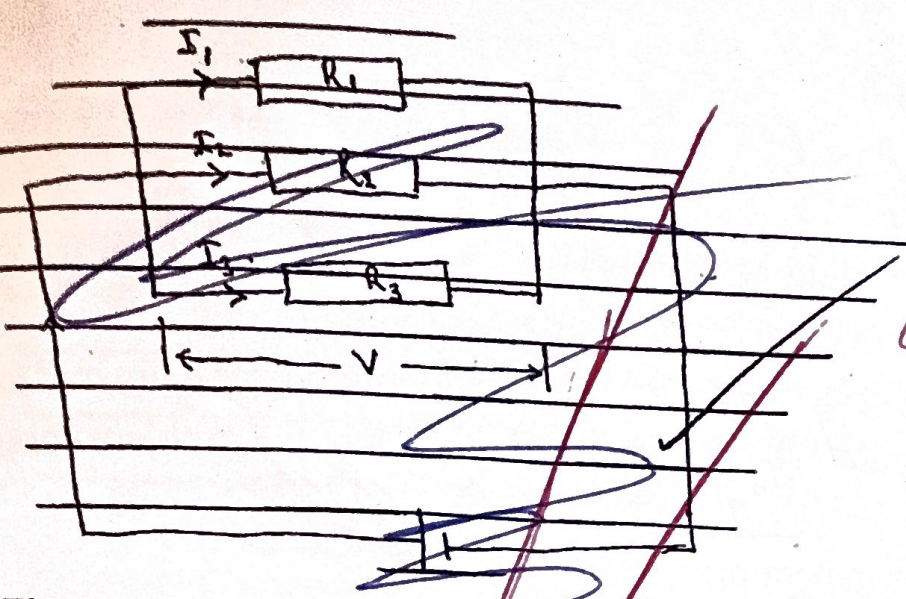
$= 1.33 \times 10^{-11} C.$

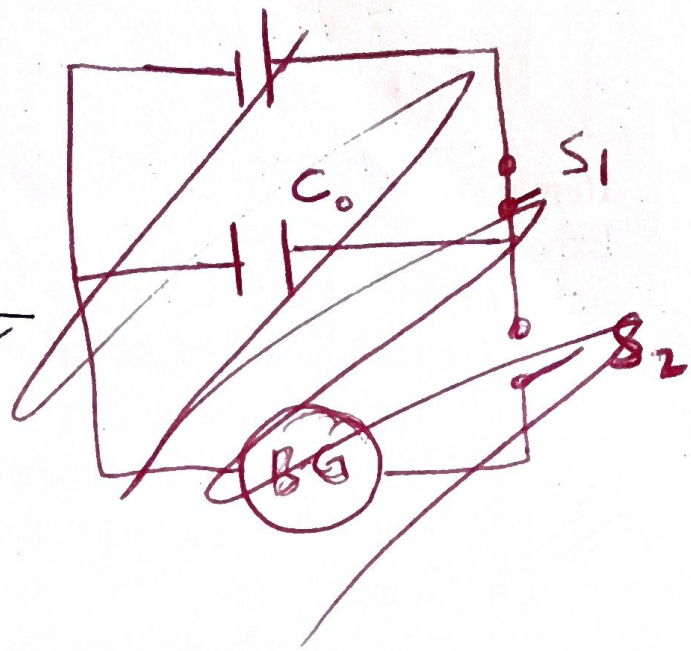
$Q = CV$

$= 2.21 \times 10^{-12} \times 6$

$= 1.33 \times 10^{-11} C$







The current I due to the South divides at the junction A such that different currents I, I and I_2 flow through resistance R_1, R_2 and R_3 respectively.

$$I = I_1 + I_2 + I_3$$

Where $I_1 = \frac{v}{R_1}$, $I_2 = \frac{v}{R_2}$ and $I_3 = \frac{v}{R_3}$.

$$I = \frac{v}{R_1} + \frac{v}{R_2} + \frac{v}{R_3}$$

$$\frac{I}{v} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{I}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_2}$$

R = equivalent resistance.

Power generated = IE But $I = \frac{E}{R+r}$

$$P = \frac{E^2}{R+r}$$

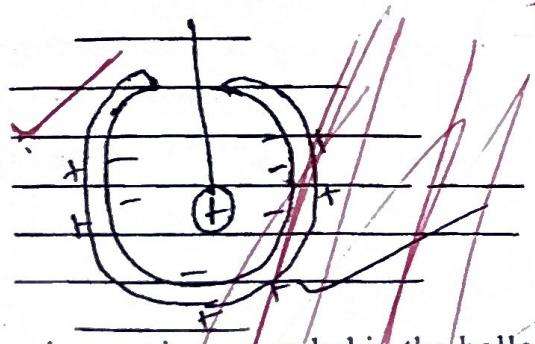
$$= \frac{20^2}{10.0+4.0}$$

P = 28.57W

Efficiency $= \frac{I^2R}{IE} = \frac{R}{R+r} \times 100\%$

$$= \frac{10}{14}$$

$$= 71.4\%$$



The delicate equipment is suspended in the hollow conductor.
Equal and opposite charges are indicated on the inner wall of the hollow condition.
External charges only affects charges on the outside surface.

Total 25

Item 6

Expected Responses

(a) (i)

The inefficiency of the pump could be because of;

- Source limitation i.e generate providing insufficient voltage.
- High impedance limiting effective current flow reducing power delivered to the motor. So we used to calculate the two and find out where the problem comes from.

Maximum induced emf by generator;

$E_0 = NBAw$

$= NBA\ .2\pi f$

$= 500 \times 0.30 \times 0.020 \times 2\pi \times 50$

$E_0 = 942.5V$

$Vms = \frac{E_0}{\sqrt{2}} = \frac{942.5}{\sqrt{2}}$

$= 666.4V$

From our calculation; the generator provides sufficient voltage so the inefficiency could be because of the impedance.

Reatances

Inductive reactances $X_L = 2\pi fL$

$= 2\pi \times 50 \times 0.60$

$= 188.5\Omega$

Capacitive eactance, $X_c = \frac{1}{2\pi(50)(60 \times 10^{-6})}$

$= 53.1\Omega$

Since inductive and capacitive reactance oppose each other in phase.

Net reactance, $X = X_L - X_c$

$= 188.5 - 53.1$

$= 135\ .4\Omega.$

Impedence

$Z = \sqrt{R^2 + x^2}$

$Z = \sqrt{20^2 + x135.4^2}$

$Z = 137\Omega$ (136.9Ω)

The current in the circuit is therefore

$I = \frac{V}{Z} = \frac{666.4}{137} = 4.86x$

If the inductor and capacitor were absent, the current would be

$I = \frac{V}{R} = \frac{666.4}{20} = 33.32x$

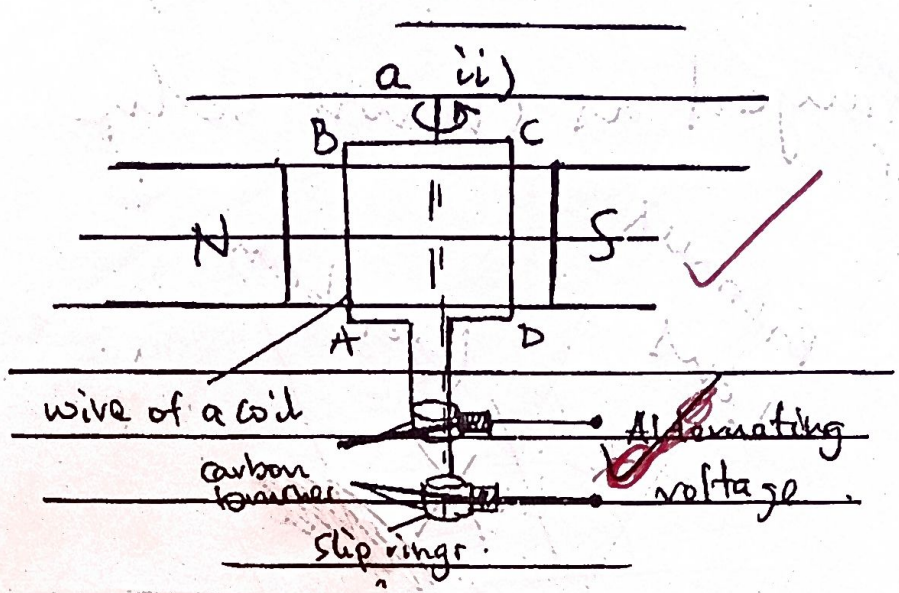
therefore impedance reduces current drawn by the motor from the source leading to inefficiency.

Improvement of efficiency

- Add a capacitor in series to increase capacitive reactance which reduces impedance hence ↑se in current. try 1
- Use of thick copper wires of low resistance to reduce unknown resistance. This reduces power loss due to heating.

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A.C generator

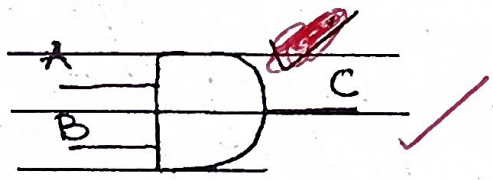


The coil is connected to an external circuit via slip rings and brushes.

The electrical energy in the system is produced by a rotating – coil generator driven by a diesel engine. As the coil rotates in the uniform magnetic field, the magnetic flux linking the coil changes continuously. This changing flux induces an emf in the coil where magnitude depends on the number of turns of the coil, the strength of the magnetic field, area of the coil and angular speed of rotation.

04

(b) The required logic gate is the AND gate which uses two inputs and one output.



Where A represents "water level" input i.e how water level is 1
High water level is 0 and B represented "Voltage level or current level" input
i.e safe is 1
unsafe is 0

04

Truth table

A	B	C
0	0	0
1	0	0
0	1	0
1	1	1

Conclusion

The pump will only be turned on when water level is low and voltage or current levels are safe.

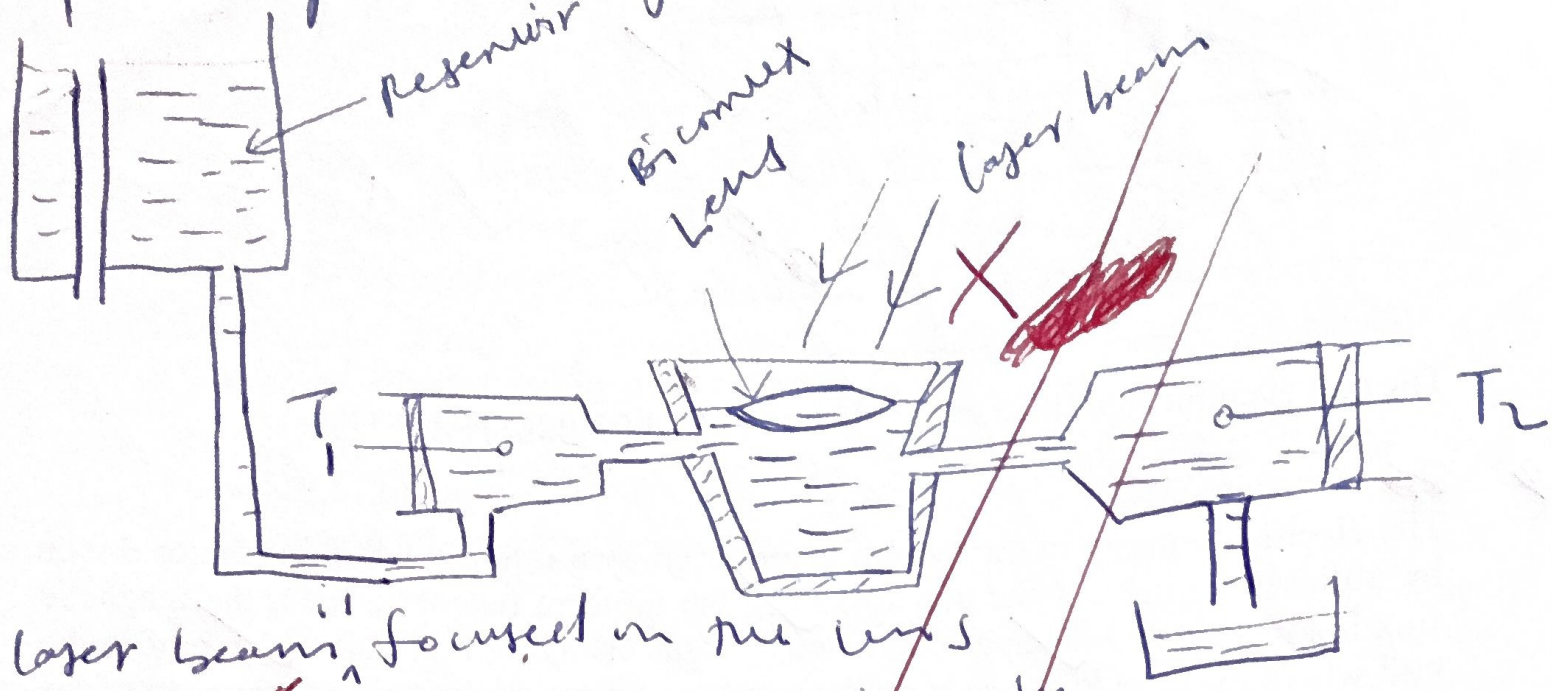
Total = 25

END

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Item 3 (continuous flow method)

- The set up is arranged as shown below.



- A laser beam is focused on the lens continuously such that it continuously absorbs heat.
- The cooling oil with known S.H.C, C_{oil} is passed around the lens at a constant flow rate.
- When steady state is achieved, the inlet temperature T_1 and outlet temperature T_2 are measured and recorded.
- A known mass, m of the cooling oil that passes through the chamber is collected in time t.
- The amount of heat Q absorbed by the lens in every second is determined from the expression

$$Q = \left(\frac{m}{t}\right) C_{oil} (T_2 - T_1)$$

- The heating coil is replaced with the lens having a laser beam continuously incident on it through the glass.
- The heat absorbed by the lens is calculated from;

$$Q = m C_{oil} (T_2 - T_1)$$

where m - mass rate of oil at that instant.
C_{oil} - S.H.C of oil obtained using the known temperatures.
T_1, T_2, inlet and outlet temperatures.